

LECTURE 5

Scenario

To eliminate peak-hour traffic bottlenecks in Hyderabad's tech corridor, the city constructed major multi-lane elevated corridors and flyovers under the Strategic Road Development Plan (SRDP)—such as the massive IKEA Flyover, Bio-Diversity Junction, and the Mindspace corridors. Initially, commute times dropped. However, three to four years later, traffic congestion at the down-ramps (Gachibowli, Kondapur, and Hitec City) is worse than before the flyovers were built.

- *What was the immediate, linear solution deployed by city planners, and why did it feel like a win at first?*
- *What unintended behaviors and secondary consequences looped back to break this solution after 2–3 years?*
- *What underlying structural incentives and mental models in Hyderabad are driving this repeating traffic cycle?*

What was the immediate, linear solution deployed by city planners, and why did it feel like a win at first?

***The Immediate Fix:** Building elevated, signal-free multi-lane flyovers over major junctions (e.g., Mindspace, Biodiversity) to bypass signal lights.*

***Initial Win:** In year one, point-to-point travel time decreased significantly because throughput at specific intersections temporarily increased.*

What unintended behaviors and secondary consequences looped back to break this solution after 2–3 years?

Induced Demand: Reduced commute times encouraged thousands of tech employees to buy personal cars or switch from public transport to solo commuting.

Bottleneck Displacement: The flyover didn't destroy vehicles; it just dumped double the traffic volume onto un-widened arterial down-ramps faster (e.g., exiting onto Kondapur or Khajaguda roads).

Urban Sprawl Expansion: Faster travel times incentivised real estate growth further out (e.g., Tellapur, Kokapet, Mokila), adding tens of thousands of long-distance daily car commuters back into the same corridor.

What underlying structural incentives and mental models in Hyderabad are driving this repeating traffic cycle?

Structures:

- ***Incentive Structure:** Poor last-mile metro/bus connectivity forces commuters into personal vehicles or private cabs.*
- ***Planning Structure:** Car-centric urban design (prioritizing vehicular speed over pedestrian footpaths and public transit dedicated lanes).*

Mental Models:

- *"Flyovers equal modern development" (Civic leadership belief).*
- *"Owning a car is a status symbol / public transport is inconvenient" (Commuter belief).*

BUILDING MORE FLYOVERS TO FIX CAR CONGESTION IN HYDERABAD

= **LIKE LOOSENING YOUR BELT TO FIX OBESITY** =



**MORE ROADS
BRING MORE CARS.**



**MORE SPACE
BRINGS MORE TRAVEL.**



**MORE SPACE
BRINGS MORE EATING.**



**ROOT CAUSE
REMAINS THE SAME.**



SOLVE THE ROOT CAUSE, NOT THE SYMPTOM.

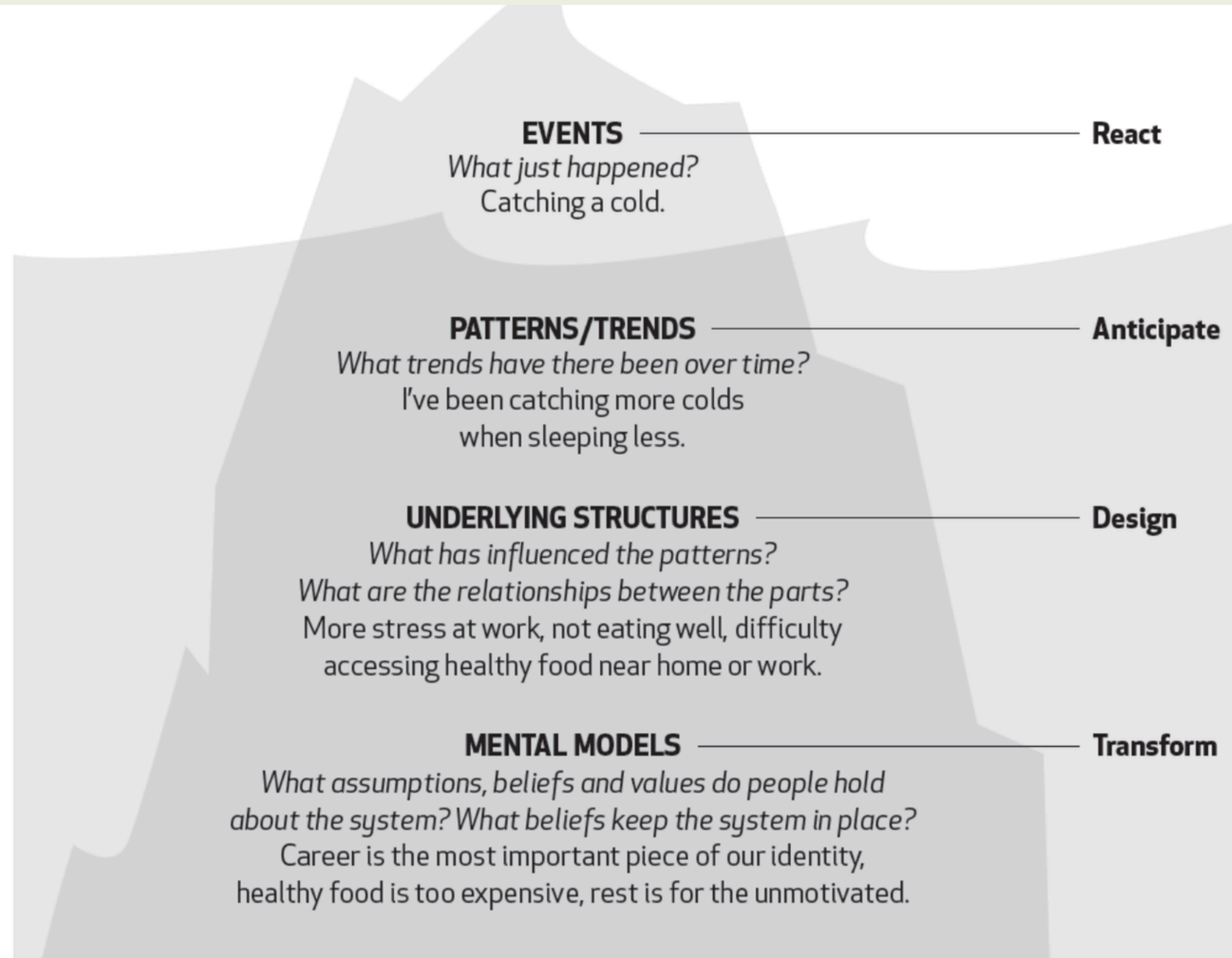
Better public transport, land use planning, walkable cities, behavior change — Healthier habits, not a looser belt.

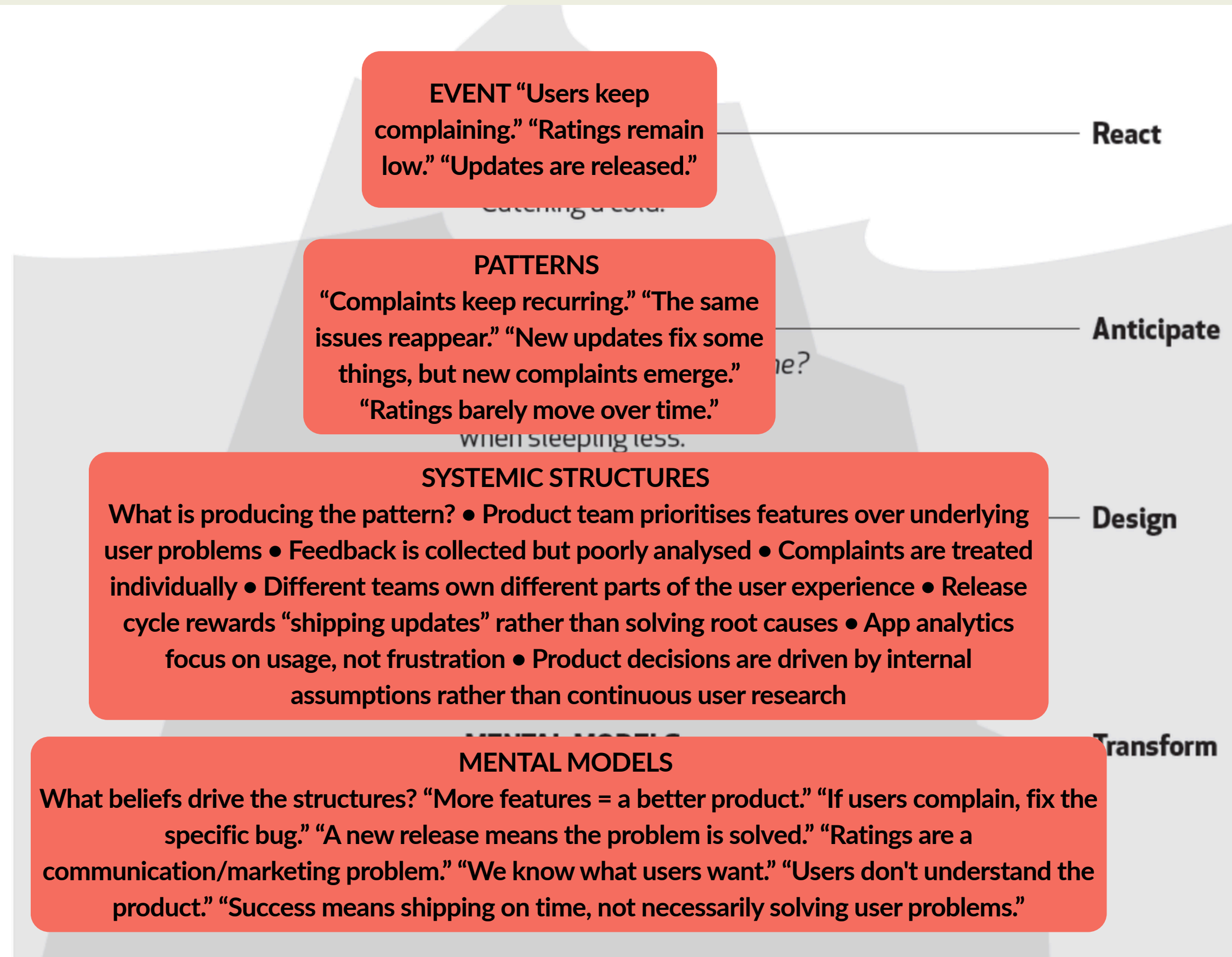
Activity 2

20 Minutes

Split into groups of 3–4. Pick one of these events

- "The coffee shop line is always 15 minutes long at 8am."
- "Our team keeps missing the same deadline every quarter."
- "Traffic on the highway is terrible every weekday evening."
- "*This app's users keep complaining, but the ratings don't improve even after updates.*"





Iceberg level	Question	Example
EVENT	What happened?	Users complain and ratings remain low.
PATTERN	What keeps happening?	Complaints recur despite repeated updates.
SYSTEMIC STRUCTURE	What produces this pattern?	Feedback, prioritisation, development and measurement systems favour quick fixes over root-cause resolution.
MENTAL MODEL	What beliefs sustain the structure?	“Shipping an update means solving the problem.”

**“Don't just fix what users complain about.
*Understand why your system keeps
giving them something to complain
about.*”**

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- "Our team keeps missing the same deadline every quarter."
- "Traffic on the highway is terrible every weekday evening."

Each group works up the iceberg from the bottom of the visible event downward, filling in:

- **Pattern:** Has this happened before? How often? Is it getting better or worse?
- **Structure:** What rules, incentives, capacity limits, or relationships are producing this pattern?
- **Mental Model:** What belief or assumption (held by the people inside the system) keeps that structure in place?

The way we think shapes our understanding about the world and informs our choices. To become more knowledgeable and improve our learning process, we can create a systems thinking mindset that enables us to recognise the complexity and interconnectedness of everything around us.

What is systems thinking?

Systems thinking involves expanding our awareness to see relationships between the parts that make up a larger entity.

This mindset includes looking at the bigger picture, considering multiple perspectives, examining how things relate, finding root causes, and challenging our assumptions.

When we use systems thinking, we are exploring ways to relate to the people and world around us. We are also anticipating future impacts and creating solutions for the systems we have put in place. Systems thinking helps us navigate risks, overcome barriers, and foster opportunities.